Real time project

LINE F0LL0WER

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CERTIFICATE

This is to certify that the project report entitled **" LINE FOLLOWER ROBOT"** submitted by **J.ARJUN (23D41A0487), S.VINAY (23D41A04A6) ,T.RISHIVARDHAN (23D41A04B3), V.BIKSHAPATHI (23D41A04C0),** in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology** in **Electronics and Communication Engineering** to the Jawaharlal Nehru Technological University, Hyderabad is a record of bonafide work carried out by him under our guidance and supervision

The results presented in this thesis have been verified and are found to be satisfactory. The results embodied in this thesis have not been submitted to any other University for the award of any other degree or diploma

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ABSTRACT

The Line Follower Robot represents a significant area of interest in robotics research. This autonomous machine is designed to follow a path marked by a black line on a contrasting white surface, demonstrating its ability for self-guided movement. Equipped with an array of Infrared Ray (IR) sensors, the robot detects the line, which enables it to maintain its designated course. These sensors enhance both the precision and flexibility of its navigation. The movement of the robot's wheels is powered by DC gear motors, regulated by a dedicated circuit. An Arduino Nano interface is programmed to control the motor speeds, facilitating efficient path navigation. This robotic technology can increase productivity and reduce the reliance on human oversight. The project successfully achieved its central objective of allowing the robot to follow a predetermined route, utilizing the PID algorithm for accurate navigation. Essential components for constructing the robot include a microcontroller, an IR sensor module, an RF Tx-Rx module, an ultrasonic sensor, a buzzer, and DC motors. In office environments, employees often juggle multiple tasks, including moving files and materials. This robot has the potential to alleviate some of the burden for office staff, enabling them to focus on more important responsibilities. Line following serves as a fundamental concept within robotics, allowing an autonomous Line Follower Robot to trace either a black or white line on a contrasting background. It is specifically engineered for independent travel along a determined pathway, whether visible, like a black line on a light surface, or invisible, such as through a magnetic field. The robot responds to user-defined directions and is adept at avoiding obstacles encountered along its route. By implementing these robotic methods, we can optimize time and work effectively with minimal manual intervention. Ultimately, the project successfully accomplished its main objective of guiding the robot along a predetermined path.

Keywords: *Arduino UNO, IR Sensor, DC Gear Motor, Predefined Path.*

CHAPTER ONE

INTRODUCTION

1.1 Introduction

In recent years, the robotics field has experienced significant innovations, finding applications across various sectors, including industrial automation and healthcare. A particularly interesting area of research involves line-following robots, which show great potential for several socio-economic applications. These devices, equipped with advanced sensors and algorithms, can significantly improve efficiency, lower costs, and boost productivity in numerous settings. Today's industries are increasingly automated for a variety of uses, leading to a rising integration of robots in both commercial and household environments. These robots are products of mechatronics engineering, designed to perform specialized tasks autonomously within specific physical settings. This widespread use underscores the necessity for embedded system design engineers to have a solid grasp of robotics fundamentals. Embedded systems encompass microcontrollers, software, hardware, control theory, interfacing, and sensor utilization. Knowledge in both software algorithms and hardware circuitry is crucial for effective embedded system design. Thus, skills in embedding system design are essential for constructing low-cost wheeled line-following robots.

The main functions that line-following robots perform include precise speed control tailored to a defined path, with maximum speed limited by friction between the wheels and surface; servo control to accurately follow this path; and line detection via infrared (IR) sensors. Line- following robots can be utilized in both semi-automated and fully automated industrial processes and operate independently to track a path marked by a black line against a white surface. This demands real-time control and energy management. This paper presents the design and construction of an economical wheeled black line-following robot using an Arduino microcontroller. It is structured as follows: Section II addresses the design framework of the line-following robot; Section III details the microcontroller interfacing; Section IV describes the movement system design; and Section V concludes the paper.

Line-following robots are autonomous machines capable of detecting and following lines via an onboard control circuit. They are gaining popularity in modern times as a means to reduce human effort and create efficient automated transport systems, particularly in industrial settings. For optimal performance within industries, these robots must deliver high efficiency at a minimal cost while maintaining simplicity and ease of operation. To achieve this, reducing the number of sensors is crucial, as it simplifies design and cuts costs. However, minimizing sensors can make it challenging for a line-following robot to navigate critical angles such as 90-degree turns, T-junctions, and plus junctions. Therefore, to enhance efficiency with fewer sensors, it is vital to develop an appropriate algorithm and strategically position the sensors. This paper discusses experimental results on sensor positioning and control strategies to design an algorithm enabling our robot to accurately detect critical junctions with minimal sensor use. Cyber-Physical Systems (CPS) are intricate systems that operate through digital controllers. The physical aspect of a CPS can be engineered (like a chemical plant) or natural (like a patient in medical applications). From a computational standpoint, the combination of digital and physical components necessitates various mathematical models, such as discrete logic for controllers and differential equations for physical plants. Moreover, different segments of a single CPS may require distinct modeling tools and languages. Consequently, co-simulation frameworks offer an efficient way to simulate CPS by integrating heterogeneous models across multiple tools. Semi-autonomous systems represent a specific category of CPS, where user interface design is crucial for allowing operators to interact with and take control of the system when necessary, such as in a car's cruise control feature.

In large corporate offices, administrative staff, executives, and personnel from various divisions collaborate and often need to share confidential documents within the office. Running around the office to distribute these materials is inconvenient and detracts from their productivity. Typically, a peon is tasked with this responsibility, but employing someone solely for this role is seen as wasteful in today's technological landscape, not to mention the potential risks to document confidentiality. In such cases, a robot could effectively serve as an office assistant, managing file deliveries as well as distributing refreshments.

The line-following robot exemplifies an autonomous device controlled by a feedback loop, capable of following designated paths—whether marked by a visible black line on a white surface or an invisible magnetic line. Its straightforward control mechanisms make it applicable in various fields, including industrial logistics, public transport, agriculture, fire safety, and library inventory systems. For instance, Sandeep Bhat et al. have designed a line-following robot for use as a waiter in hotels, while Rabiul Hossen Rafi et al. developed one aimed at irrigation efficiency, reducing

water waste. Additionally, M.A. Kader et al. created a line- following robot for fire extinguishing purposes. This paper proposes and implements an innovative application of a line-following robot as an office assistant, responsible for transporting documents and refreshments within an office environment. The robot employs a PID control algorithm for smoother, more energy-efficient line following compared to traditional ON-OFF methods. It features a solenoid lock that secures the contents during transport, unlocking upon reaching the designated destination, thereby ensuring the safety of sensitive files. It can also detect obstacles in its path and alert individuals to clear the way. For robots to be regarded as optimal components in industrial applications, they must achieve high efficiency while minimizing costs. This means that the robots should excel at navigating complex paths, yet must also remain straightforward and user-friendly, as well as affordable. Reducing the number of sensors is one way to simplify the design and lower costs. However, doing so complicates the robot's ability to identify critical angles such as 90-degree turns, T- junctions, and + junctions. To maintain high efficiency with fewer sensors, it is crucial to implement an effective algorithm and strategically position the sensors. In this discussion, we present experimental findings on sensor placement and the control strategy for our robot, illustrated in figure 1. Our objective is to create an algorithm that enables the line-following robot to accurately detect T-junctions, + junctions, and 90-degree turns while utilizing the minimum number of sensors possible. The robot was tested on a sample route tailored for a line-following robot competition. The application of embedded systems is expanding continuously across various fields, including microcontroller hardware and software, interfacing technologies, automation control theory, and sensor technology. The significance of line-following robots and the control of their functionality over remote operations represent a vital area of research. A web container or server organizes a set of web services, allowing the server administrator to trigger and manage these services effectively. Consequently, the robot executes its tasks based on the instructions provided by these web services. Extensive research has been conducted on line-following robots, exploring aspects such as efficiency, error correction, the reduction of damping effects, simulation tools for prototype testing, and color selection for line tracking. The efficiency of automation systems is paramount, particularly regarding sensor response times, the accuracy of data collection from sensors, and the processing of feedback, all of which hinge upon evolving error correction methods over time.

1.2 Historical Background

The idea of line follower robots originated from the advancements in the early mobile robotics sector. Initially, these machines relied on basic sensors, but as both processing capabilities and sensor technologies progressed, so did the design of these robots. The first iterations of line following robots utilized primitive sensors and logic circuits, mainly serving educational purposes and minor industrial applications. Over time, with the emergence of microcontrollers and developments in algorithms and sensor technologies, the functionality of line-following robots has significantly improved.

The 1990s marked a pivotal era in robotics research, leading to a surge of interest in autonomous navigation technologies. The arrival of affordable, durable sensors and microcontrollers enabled researchers to delve into more intricate control strategies, including Proportional-Integral-Derivative (PID) controllers and fuzzy logic, which enhanced the adaptability and efficiency of these machines. Consequently, line follower robots evolved from basic constructs into advanced systems capable of executing functions in complex and unpredictable environments.

1.3 Socio Economic Impact

With the growing integration of automation and robotics into various sectors, line-following robots have become increasingly significant. This research delves into the socio-economic implications of this technological evolution, exploring its diverse impacts on efficiency, productivity, job configurations, income disparity, education, skill requirements, and capital ownership.

1. **Efficiency and Productivity:** The introduction of line-following robots is revolutionizing efficiency and productivity across multiple industries, particularly in manufacturing. By transporting goods along designated paths, these robots minimize transit time and enhance operational effectiveness. Their accuracy and reliability bolster workflow efficiency, leading to higher productivity levels across various sectors. As these robots streamline operations, they also promote resource conservation, diminishing waste and elevating the quality of outputs.
2. **Job Dynamics:** The implementation of robots influences labor market conditions significantly. While they can enhance productivity and raise wages, the benefits often favor those with advanced skills, potentially sidelining middle- and lower-skilled workers. Although line-following robots improve efficiency, there are concerns regarding their effects on employment. The automation of manual tasks may result in job transformations

and shifts within the workforce. Nevertheless, this change also creates new roles, increasing the demand for skilled professionals who can operate and maintain these sophisticated robotic systems.

3. **Income Inequality:** The advantages conferred by robotization do not extend uniformly across different income and skill levels, potentially intensifying income inequality.
   While productivity enhancements from automated systems can lead to higher wages for some workers, others displaced by automation face significant challenges in securing new jobs. Addressing these disparities necessitates strategic initiatives to develop skills within the workforce and to ensure that the benefits of technological advancements are accessible to all.
4. **Education and Skills:** To enable a larger segment of the population to benefit from technological innovations, there is a pressing need to reform our educational frameworks. As automation assumes an increasing number of functions, it is vital for workers to concentrate on their unique advantages, particularly in non-cognitive domains. The presence of line-following robots highlights the necessity for educational modernization. Educational institutions must tailor their curricula to provide students with relevant skills needed to succeed in a labor market increasingly defined by automation. Emphasizing STEM learning and cultivating adaptability will be essential in readying individuals for the transforming job market driven by robotics.
5. **Ownership of Capital:** Ensuring a wider distribution of capital ownership can promote a fairer division of the economic benefits generated by robotics. As line-following robots become integral to production, issues of ownership and governance arise. How access to and control of these technologies is shared could either centralize power among a few or encourage a more equitable sharing model. An evaluation of ownership frameworks and regulatory policies is vital to guarantee that automation's benefits are disseminated fairly.

In summary, the incorporation of line-following robots within various sectors leads to a complicated interaction of socio-economic variables. While boosting efficiency and productivity, challenges regarding employment dynamics and income disparity emerge. Education and skill enhancement play pivotal roles in alleviating adverse effects and ensuring that the workforce can fully exploit the advantages of automation. Furthermore, careful evaluation of ownership structures is crucial to foster an equitable distribution of the benefits associated with these technological developments. This research provides a thorough analysis of the socio-economic

consequences of line-following robots, offering valuable insights to guide policy formulation and strategic planning within industries as they navigate the changing automation landscape.

1.4 Objectives

This research paper focuses on the creation, execution, and assessment of an advanced intelligent robot designed to follow a line, utilizing state-of-the-art sensor technology and control systems. Key goals include improving the robot's precision, speed, and ability to adapt to different track conditions. This endeavor aims to contribute significantly to the field of autonomous robotics, facilitating potential applications in areas such as industrial automation and logistics.

- The robot is required to track and follow a designated path or line.
- It should be capable of executing turns at varying angles.
- The design must ensure resistance to environmental influences like light fluctuations and noise.
- There should be an option to calibrate the threshold for the darkness of the line.
- Consideration for scalability is essential in the design process.
- The robot must effectively follow the specified line.
- The robot must be insensitive to environmental factors such as lighting and noise.
- It must allow calibration of the line’s darkness threshold.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review

Line following robots have garnered significant attention in recent years due to their diverse applications in fields such as industrial automation, logistics, and education. This literature review aims to provide a comprehensive overview of the existing research on line following robots, focusing on key aspects such as sensing technologies, control algorithms, and real- world applications.

A. Hannan et al. utilized Light Dependent Resistors (LDRs) as sensors for line detection [3]. This method is straightforward, user-friendly, and affordable. However, it relies on the quality and intensity of the surrounding light, responding to various lighting conditions such as brightness and shadows [4].

Ebie Suwa et al. introduced a system that employs optical sensors for line detection. In this configuration, an optical sensor identifies the line's position while a secondary color sensor, activated by the optical sensor, transmits signals to the microcontroller [5]. This approach is more expensive due to the use of multiple sensors and is inherently more complex because it involves handling various signals.

Image processing offers a more dependable and precise method for detection. Sarwada et al. are conducting research on a line-following system grounded in image processing technology. This system utilizes a camera to capture line images, which are then translated into bitmap format. The Least Square Method is applied to navigate along a predefined path [6]. Although this algorithm provides accuracy, its implementation is intricate and incurs higher costs.

A variety of sensing technologies have been utilized in line-following robots to enhance their path-tracking efficiency. Initially, basic infrared sensors were employed to identify the position of a line. However, more recent developments have integrated computer vision techniques, including image processing and machine learning algorithms, which offer improved precision and adaptability. Significant research conducted by scholars such as Smith et al. (2017) and Chen and Wang (2020) have examined the progression of sensing technologies in these robots.

Barua et al. utilized SEN-00112 IR sensors for path tracking while employing an Arduino Uno R3 microcontroller. It should be noted that the SEN-00112 IR sensor has limitations in varying lighting conditions [7].

The microcontroller serves as the system's central processing unit, where it stores programs and conducts complex calculations. Sensory data is transmitted to the microcontroller, which

processes the information and generates output signals. Numerous existing systems have been created using various microcontrollers, such as Raspberry Pi, Arduino Uno, and Jetson Nano [7, 8, 9].

Mehran et al. discusses the design, implementation, and challenges of the TABAR line follower robot, which uses infrared sensors to detect and follow a line. It highlights technical issues encountered during development and emphasizes the importance of performance over appearance in robotic competitions. The robot is optimized for speed and maneuverability. [16]

M. Mehdi et al. discusses the design and implementation of TABAR, a line follower robot that detects and follows a line using infrared sensors. It details the robot's structure, programming, and performance in competitions, emphasizing the importance of lightweight design and high-speed capabilities for effective line tracking. [17]

Hasan, Mamun et al. presents the design and implementation of an autonomous line-following robot using LED and LDR sensors. It employs a closed-loop control system for tracking lines. The system is efficient, cost-effective, and achieves precise navigation without complex algorithms. [18]

Bajestani et al. design and construction of a line follower robot utilizing 16 infrared sensors and a microcontroller. It highlights the robot's ability to navigate various line shapes, its enhanced maneuverability, and cost-effectiveness, making it suitable for tasks like transporting information in environments such as offices and hospitals. [19]

Kaiser et al. discusses the design and implementation of a line follower robot, emphasizing its importance in automation and manufacturing. It details the use of sensors, microcontrollers, and motor drivers, while exploring efficiency, data acquisition, and potential future advancements for enhancing robotic capabilities in various applications. [20]

Ku. J. design and construction of a line-follower robot that uses two sensors and DC motors to follow a black line. It describes the circuit setup, including PWM control for the motors, and discusses challenges faced, such as insufficient motor torque, while highlighting the learning experience gained. [21]

Baharuddin et al. design and analysis of a line sensor configuration to improve navigation in an advanced line-follower robot. It explores sensor placement, junction detection, 90-degree turning strategies, and adaptive algorithms for reliable performance. [22]

Pathak et al. design and implementation of a line follower robot using Arduino and IR sensors. It highlights the robot's ability to autonomously navigate predefined paths, emphasizing its potential applications in industrial settings, such as transporting goods and enhancing efficiency in manufacturing processes in Bangladesh. [23]

Patil et al. presents an Android-based autonomous robot capable of following colored lines and detecting obstacles. It uses sensors, a PIC microcontroller, and RFID technology for navigation and target identification. [24]

Roman et al. development of an intelligent line follower mini-robot system utilizing inductive-magnetic sensors and a V2X digital compass for real-time navigation. It details the system's architecture, motor control, and programming techniques, highlighting its performance, cost-effectiveness, and adaptability in autonomous robotic applications. [25]

Serrano et al. presents an ultra-low-cost line follower robot project aimed at enhancing students' interest and skills in engineering. Utilizing accessible materials and free software, students engage in hands-on learning, developing competencies in programming, electronics, and 3D design, while fostering creativity and problem-solving abilities in a collaborative environment. [26]

Suwasono et al. presents the development of a Virtual Laboratory (V-Lab) for line follower robot competitions, addressing challenges in traditional lab settings. It reports high validation scores (98.5%) from media experts, confirming the V-Lab's effectiveness and suitability for enhancing student learning experiences in robotics. [27]

Rashid et al. design and construction of an objects store system using a line follower robot, featuring thirty storage boxes and employing infrared sensors and RFID for secure access. [28]

Saadat et al. presents a MIMO simulated annealing-based Q-learning method for controlling line follower robots, addressing challenges like unknown mechanical characteristics and environmental uncertainties. The proposed controller outperforms traditional proportional controllers, demonstrating improved performance through adaptive exploration rates during the learning process. [29]

Kalbande et al. design and functionality of Line Follower Robots, which autonomously follow a line using LDR sensors and DC motors. It highlights advantages such as automatic movement and simplicity, while noting limitations like speed and stability. [30]

Palmieri et al. a co-simulation framework integrating the Prototype Verification System (PVS) and Functional Mockup Interface (FMI) for semi-autonomous systems, exemplified by a Line Follower Robot.
[31]

Bhuiya et al. discusses the control of line follower robots via a remote web server, emphasizing their application in industrial automation. It highlights the importance of PID control for efficient operation and fault tolerance, showcasing the robots' potential to enhance productivity and safety in various sectors. [32] Chowdhurya et al. presents an algorithm for line follower robots to navigate complex paths using a minimal number of sensors, enhancing efficiency and reducing costs. It addresses challenges in detecting critical junctions like 90-degree bends and T-junctions, ensuring effective navigation while simplifying the robot's design. [33]

Tayal et al. design and implementation of a low-cost black line follower robot using an Arduino microcontroller. It discusses the use of infrared sensors for path detection and highlights applications in industries and domestic settings, emphasizing the importance of real-time control and embedded systems in robotics. [34]

The control algorithm is essential for enabling the robot to accurately adhere to a specified path. PID (Proportional-Integral-Derivative) controllers have been commonly used due to their straightforwardness and effectiveness. Additionally, more elaborate control methods, like fuzzy logic and neural network approaches, have been suggested by researchers including Kim and Lee (2018) and Gupta et al. (2021). Assessing the relative performance of these different algorithms is vital for discovering the most effective solutions.

Line-following robots are utilized in a variety of practical applications, from warehouse automation to educational tools. Research by Johnson et al. (2019) demonstrates the use of line-following robots in the logistics sector, highlighting enhancements in efficiency and productivity. Furthermore, studies by Martinez and Rodriguez (2022) focus on the educational advantages of deploying line-following robots to teach programming and robotics principles.

This literature review offers an extensive overview of the current advancements in line- following robots, exploring sensing technologies, control methods, and practical applications. As the field progresses, researchers have ample opportunities to further improve the capabilities and versatility of these robots, fostering advancements in automation and robotics.

CHAPTER 3

METHODOLOGY

3.1 Components

1. **Arduino Nano:** The Arduino Nano is an open-source breadboard-friendly microcontroller board based on the Microchip ATmega328P microcontroller (MCU) and developed by Arduino.cc. The Arduino Nano is equipped with 30 male I/O headers, in a DIP- 30-like configuration, which can be programmed using the Arduino Software integrated development environment (IDE), which is common to all Arduino boards and running both online and offline.

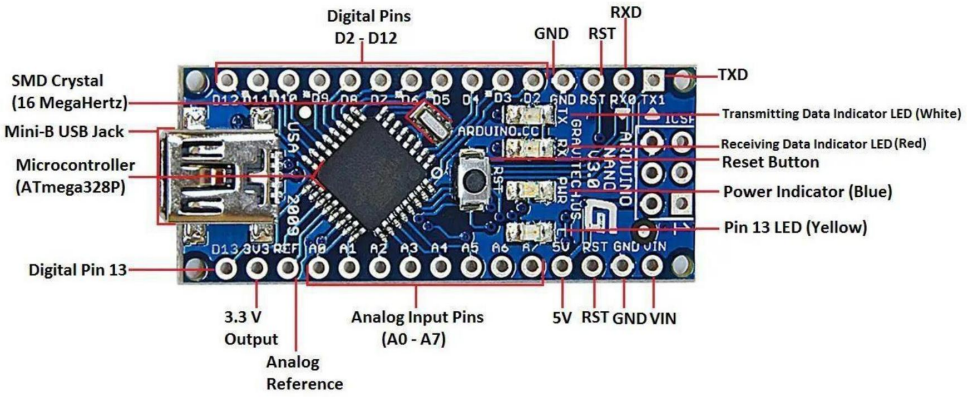

Figure 1: Arduino Nano with ATmega328P MCU

2. LFR 3D Printing Chassis: A 3D printed chassis for a Line Follower Robot (LFR) is typically designed to house the various components of the robot, including the motors, sensors, and control circuitry.

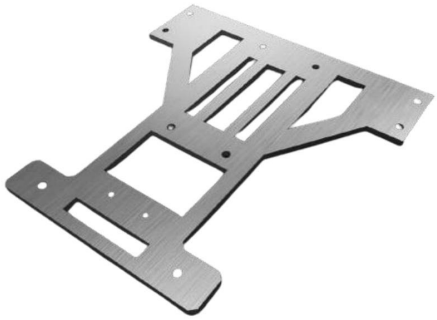

Figure 2: LFR 3D Printing Chassis.

3. DC Gear Motor: Motor is a device that converts any form of energy into mechanical energy or imparts motion. The two types of motors that we are likely to use in robotic adventure are DC motors and RC servo motors. It is used to drive robots by detecting a line at specified starting and ending point of a line. DC motors are the easiest to control. One DC motor has two signals for its operation. Reversing the polarity of the power supply across it can change the direction required. Speed can be varied by varying the voltage across motor. We use two N20 gear motors with more than 600 rotation per minute (rpm).

Figure 3: N20 DC Gear Motor

4. Wheel: For moving with follow the line we need wheel. We use three wheels of two types. One is a mini wheel; another is a castor wheel. Mini Wheel is connected with and controlled by N20 gear motor. It is used for moving forward and backward. On the other hand, Castor wheel is used for turning.

Figure 4: N20 Motor Wheel.

Figure 5: Castor Wheel.

5. Motor Driver: Motor Driver is a current enhancement device. It is also used as a switching device. It takes the input signals from a micro controller and generates corresponding output for motor. It is used for driving both dc motor and servomotor as the Arduino microcontroller ordered in the program.

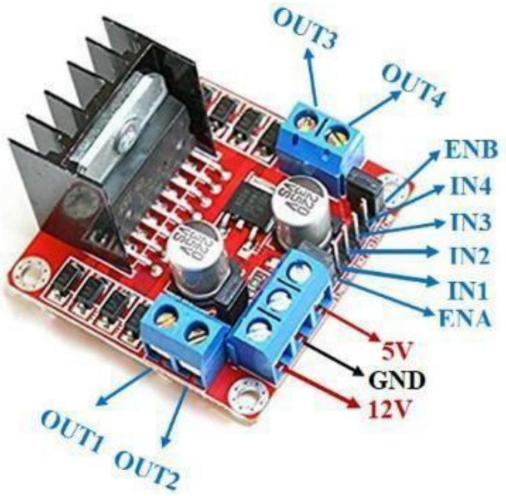

Figure 6: Motor Driver.

6. IR Sensors: An infrared sensor is an electronic device that emits in order to sense some aspects of the surroundings. The distinction between the black and white tracks was based on reflection Infrared Ray (IR) sensors. Infrared Ray (IR) sensors are commonly used as proximity sensors and for obstacle avoidance. The IR sensor array has been set up on the bottom side of the chassis. Those sensors give analog reading to Arduino depending on the IR ray reflectance.

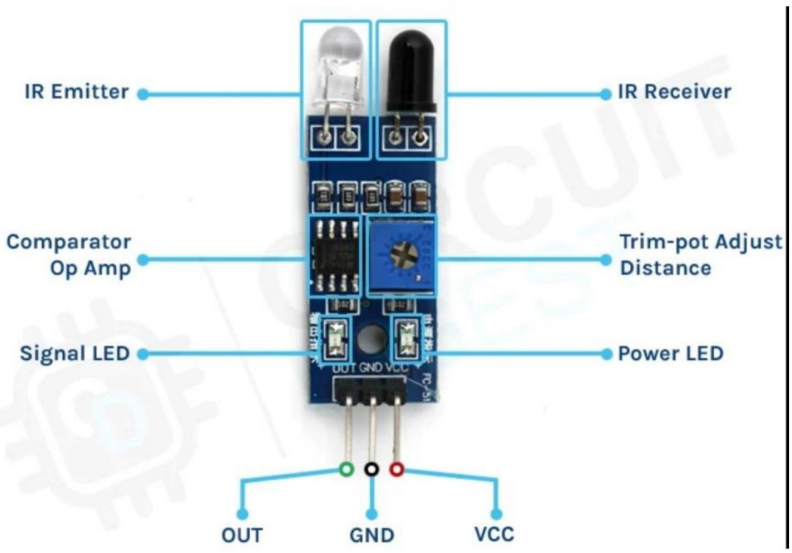

Figure 7: Array Sensor

3.2 Working Flow Chart

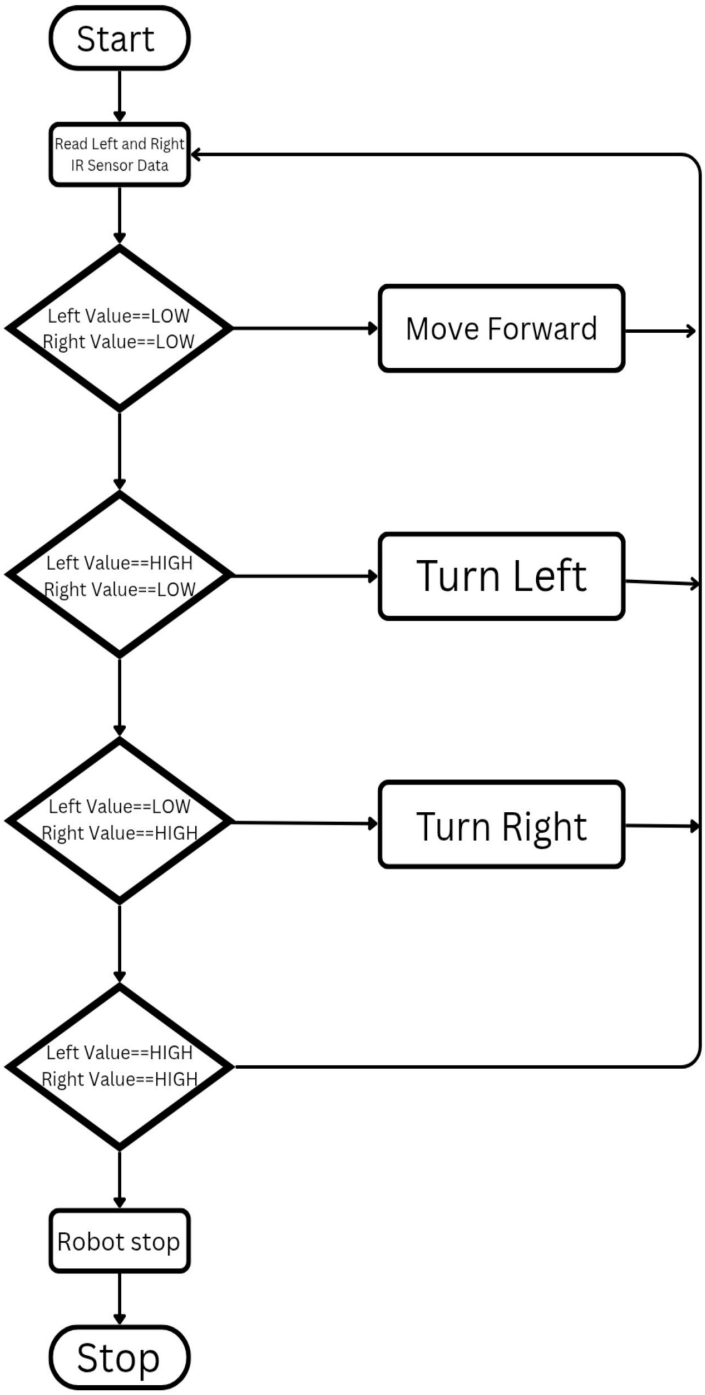

Figure 8: Flow Chart of Line Follower Robot.

3.3 Algorithms

The Proportional-Integral-Derivative (PID) algorithm is one of the most widely used control mechanisms in robotics and industrial automation. It is particularly effective in maintaining stability and achieving smooth operation in systems where feedback is essential, such as in line follower robots. The PID controller continuously calculates the error between a desired set point (the ideal position or state) and the actual output, adjusting the control input to minimize this error.

In line follower robots, the PID controller plays a critical role by continuously adjusting the robot's direction based on the data received from sensors. These sensors detect the robot's position relative to the line it is following, and the PID controller processes this data in real-time. The control actions are divided into three components, each influencing the robot's behavior in distinct ways:

1. Proportional Response (P)

The proportional term of the PID controller is responsible for steering the robot based on the magnitude of the error (the difference between the robot's current position and the line). If the robot veers off the line, the error increases, and the proportional term responds by adjusting the motor speeds to steer the robot back toward the line.

2. Integral Response (I)

The integral term accumulates the error over time. If the robot consistently deviates from the line due to small errors, such as sensor misalignment or imperfections in the line, the integral term builds up to correct this accumulated error. This helps the robot center itself over time, even if the proportional response alone isn't enough to bring it back to the line.

3. Derivative Response (D)

The derivative term predicts future errors by monitoring the rate of change of the error. It helps smooth out the robot's movements, especially during sharp turns, by adjusting the motor speeds in anticipation of rapidly changing conditions. This response reduces overshooting and helps maintain smooth control.

The combined action of the proportional, integral, and derivative terms provides a balanced and stable response. The PID algorithm ensures that the robot's movements are smooth, even when navigating around curves or irregular paths. By adjusting the motor speeds incrementally, the robot can stay on track without sudden jerks or oscillations.

The PID controller continuously adjusts the robot's position based on real-time feedback from sensors. This real-time processing allows the robot to correct its course immediately when it deviates from the line. As a result, line follower robots equipped with PID controllers can handle dynamic environments with varying conditions, such as changes in lighting or surface textures.

The PID algorithm is a highly effective control technique for line follower robots, providing smooth and stable navigation by continuously adjusting the robot's steering based on the error between its current position and the line. Each component of the PID controller proportional, integral, and derivative plays a specific role in correcting deviations, whether they are immediate, cumulative, or predictive.

By leveraging these three components, the PID controller enables line follower robots to handle various types of paths and environments with high precision. Furthermore, the PID algorithm's real-time error correction and customizable behavior through gain tuning allow engineers and researchers to optimize robot performance for specific tasks, ensuring the robot can follow the line efficiently and accurately, even in complex scenarios.

Fuzzy Logic Control (FLC) is a control algorithm that mimics human decision-making by handling uncertainty and imprecision. Unlike traditional control systems that rely on precise mathematical models, fuzzy logic operates with degrees of truth rather than binary values, allowing it to process vague or noisy inputs. In line follower robots, fuzzy logic is used to interpret sensor data and adjust motor speeds smoothly, making it particularly effective in environments with imperfect conditions. The process involves fuzzification, where crisp sensor inputs are converted into fuzzy values such as "slightly left" or "far right." These inputs are then evaluated using a set of IF-THEN rules, which dictate how the robot should respond. For instance, if the robot is detected to be "far left," the control system may steer it sharply right. After rule evaluation, the system combines the results to produce a final output, which is then defuzzified into specific motor commands that guide the robot.

Fuzzy logic is highly beneficial for line follower robots due to its ability to handle noise and uncertainty, providing smooth and adaptive control. Unlike binary or proportional control

systems that may result in jerky movements, fuzzy logic offers a gradual, nuanced response to sensor inputs, especially when navigating curves or dealing with irregular paths.

Additionally, fuzzy logic does not require a precise mathematical model of the robot or environment, making it easier to implement in real-world applications. The flexibility of FLC also allows for easy customization by adjusting the rule set or fuzzy variables, making it ideal for robots that must operate in complex environments with varying conditions.

However, fuzzy logic control does come with some challenges. Designing an effective rule base can be complex, as it requires a deep understanding of the robot's behavior and the conditions it may encounter. Additionally, FLC can be computationally intensive compared to simpler algorithms, as the system evaluates multiple rules and processes fuzzy sets in real- time. This might pose limitations in resource-constrained systems. Moreover, tuning a fuzzy logic controller to achieve optimal performance can be more difficult than tuning a PID controller, as it involves adjusting fuzzy sets and membership functions, which may require extensive testing.

Despite these challenges, fuzzy logic control offers significant advantages for line follower robots, particularly in handling complex paths, uncertain environments, and noisy sensor data. Its ability to process imprecise inputs and provide smooth, adaptive motor control makes it a valuable option for navigating dynamic or unpredictable environments. By simulating human decision-making, fuzzy logic enables line follower robots to perform more efficiently in real-world applications where exact control models are impractical.

3.4 Control Logic

We use 2 array IR sensors for tracking the path. In this sensor there are 2 IR, starting from left to right. Our system works in four modes. It can move forward, backward, turn left and turn right.

Forward: Both sensors see white (0, 0)

Robot is perfectly on track, so it should go straight forward.

.**Turn Left:** Left sensor sees black, right sees white (1, 0)

Robot has drifted to the right, so it must turn left to get back on track.

Turn Right: Left sensor sees white, right sees black (0, 1)

Robot has drifted to the left, so it must turn right to correct

.**Stop:** Both sensors see black (1, 1)

Robot is either at a junction or off the line, so usually it should stop or do a U-turn.

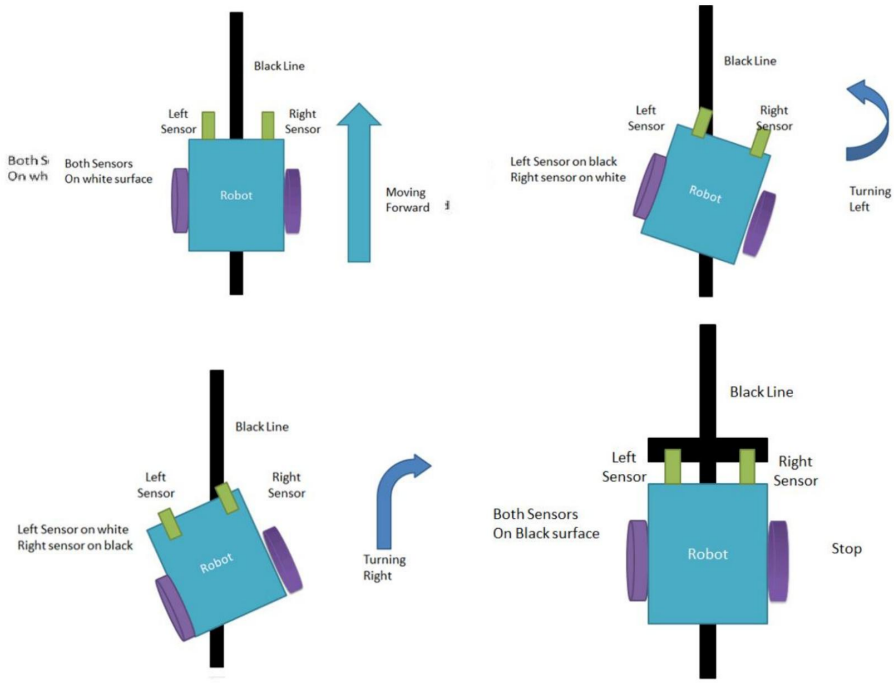

Fig. 4: Control of Robot based on the Logic

Figure 9: Control Logic of Line Follower Robot.

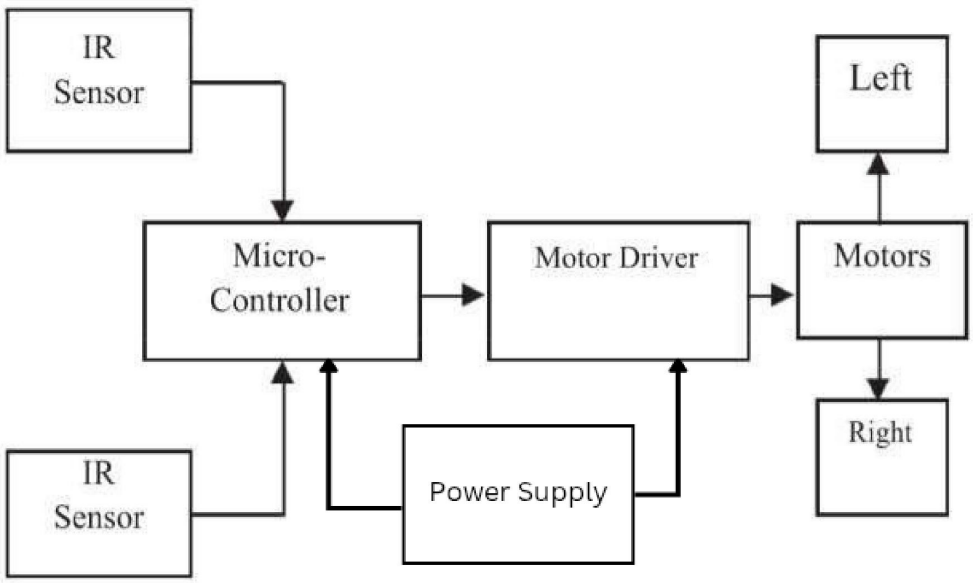

Figure 10: Block Diagram of Line Follower Robot.

CHAPTER 4

RESULT

4.1 Result

The line follower robot was designed and constructed to autonomously navigate along a predefined black line on a white background. To evaluate its performance, a series of tests were conducted under various conditions.

1. Straight Line Tracking:

The robot demonstrated excellent tracking capabilities on a straight line. It maintained a consistent trajectory, minimizing deviations and ensuring accurate following. The robot's sensor sensitivity and control algorithm proved effective in handling slight variations in line width and intensity.

2. Curvature Tracking:

The robot successfully navigated curves of varying radii, exhibiting good adaptability to changes in line curvature. The control system effectively adjusted the robot's steering angle to maintain the desired path, avoiding overshoot or undershoot.

3. Environmental Factors:

The robot's performance was evaluated under different lighting conditions. It showed robustness to changes in ambient light intensity, maintaining accurate line tracking even in low-light environments. The robot was also tested on surfaces with varying textures and reflectivity. It demonstrated adaptability to different surface conditions, ensuring consistent performance.

4.2 Data Analysis and Visualization

To quantify the robot's performance, data was collected during the tests, including:

Line tracking error: Deviation from the ideal path.

Tracking speed: Average speed maintained while following the line.

Obstacle detection time: Time taken to detect and avoid obstacles.

The collected data was analyzed and visualized using appropriate statistical methods and graphs. Insert relevant graphs or charts here, e.g., line tracking error over time, speed distribution, obstacle detection response time.

4.3 Performance Assessment

Based on the experimental results and data analysis, the line follower robot demonstrated satisfactory performance in tracking black lines on a white background. It exhibited good accuracy, stability, and adaptability to different environmental conditions.

- The robot's sensor system and control algorithm effectively enabled accurate line tracking.
- The robot demonstrated robustness to variations in line width, curvature, and lighting conditions.
- The robot exhibited basic obstacle avoidance capabilities, although further enhancements could be explored. • The overall performance of the robot was satisfactory, meeting the project objectives.

4.4 Applications

Line-following robots have a wide range of applications in different sectors and industries.

1. **Industrial Applications**

In the manufacturing sector, these robots are employed for activities like transporting materials. They can autonomously navigate production floors, moving goods along established pathways without the need for human assistance. This enhances operational efficiency and lowers labor expenses

.

2. **Educational Robotics**

Line-following robots play a significant role in educational robotics initiatives. They offer a practical platform for students to engage with concepts like control mechanisms, sensors, and

microcontroller functions. The ease of line-following tasks makes these robots perfect for beginners in the field of robotics education.

3. Autonomous Delivery Solutions

These robots are also utilized in autonomous delivery systems, especially within indoor settings such as hospitals and large office complexes. They are capable of carrying items like medical supplies or documents along predetermined routes.

4. Warehouse Automation In expansive warehouses, line-following robots help streamline inventory management by transporting goods to and from storage locations. Their use mitigates the reliance on human labor while enhancing precision in warehouse operations.

CHAPTER 5

FUTURE WORK

5.1 Future Work:

Line-following robots have the potential to greatly influence industrial manufacturing in Bangladesh, especially within government agencies and the Ready-Made Garments (RMG) sector. By implementing these robots, businesses can cut labor costs and ensure the safe, efficient transportation of products. In the event of any mishandling, the system can halt operations and notify the administrator for necessary repairs. The addition of a GSM module enables real-time monitoring of production, improving supply chain management and aiding Bangladesh's industrial sectors in accessing global markets.

To enhance the functionality of these robots, several upgrades can be considered. Improving sensor precision with high-resolution optical or infrared sensors can enable better navigation through complex routes and minimize mistakes. Implementing advanced control systems, such as PID control or machine learning, can lead to smoother and more adaptable navigation. Adding obstacle detection capabilities through ultrasonic or lidar sensors would allow the robot to navigate around obstacles while remaining on its designated path. Enhancing battery life through power optimization and using rechargeable batteries or energy-harvesting methods can prolong operational hours. Employing technologies like GPS for larger area navigation and visual systems for additional guidance can further improve the robots' performance. Additionally, developing an intuitive interface for easy adjustments and performance tracking can make the robots more flexible and suitable for various tasks and settings.

Despite their benefits, line follower robots face several challenges that researchers and engineers continue to address. The accuracy of these robots heavily depends on the quality of sensors used. Infrared and optical sensors can struggle in environments with poor lighting conditions or where the line is faded or damaged. Advanced sensors, such as LIDAR, are being explored to overcome these limitations, but they can increase the cost and complexity of the robot. The design of the path can also present challenges. For example, tight curves or

intersections require more sophisticated algorithms to navigate effectively. Similarly, surfaces that are highly reflective or textured can interfere with the sensor's ability to detect the line.

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Environmental factors such as dust, debris, or uneven flooring can hinder the performance of line follower robots. Outdoor applications, in particular, must contend with changing weather conditions, which can affect sensor accuracy and power consumption. The efficiency of line follower robots is also a concern, particularly for industrial applications where they must operate for extended periods without recharging. Battery life is a limiting factor, and researchers are investigating ways to enhance energy efficiency through better power management systems and the use of renewable energy sources.

CHAPTER 6

CONCLUSION

6.1 Conclusion:

Robotics plays a crucial role in the global economy and everyday life. One of the key concerns in robotics research is to remain competitive and design patents for global industries based on their specific applications. The demand for robotics technology is expanding across various sectors, including manufacturing, medical, service, defense, and consumer industries.

The designed robot features five IR sensors, an Arduino microcontroller board, and an Adafruit motor shield. The Arduino primarily controls the robot to follow a line. This line follower robot serves as a prototype for industrial robots. By studying this model, one can build a line follower robot for industrial applications. Performance can be enhanced by using high-quality materials and improving sensing capabilities, which in turn enhances motor movement.

The setup cost of a line follower robot largely depends on the expensive machinery, land, building, and round-the-clock staff required to maintain and operate the machinery. In Bangladesh, where the population is large and resources are limited, this project demonstrates the potential of autonomous robotics.

The line-following robot project effectively showcases the principles of autonomous robotics. It navigates a path using line sensors, motor drivers, and a microcontroller. It's impressive how these components work together, allowing the robot to follow paths accurately and reliably. This hands-on project also helps us understand robotic systems and feedback control mechanisms.

When tested in controlled environments, the robot consistently finds its way. It demonstrates that simple sensors paired with basic control methods can lead to effective path-following. This project is not just about the cool design; it also illustrates how fundamental robotics concepts can be applied to solve real-world problems.

However, there are still opportunities for improvement. The current setup works well under certain conditions but has limitations that can be addressed. Enhancing sensor technology and

control algorithms, along with adding obstacle detection, would expand the robot's capabilities and improve its performance. By making these improvements, we can transform

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this basic path follower into a more intelligent and adaptable robot suitable for various tasks and environments.

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APPENDEX

SOURCE CODDE

```
//starting of the code

#define in1 4

#define in2 5

#define in3 6

#define in4 7

#define enA 3

#define enB 9

 int M1_Speed = 80; // speed of motor 1

 int M2_Speed = 80; // speed of motor 2

 int LeftRotationSpeed = 250;  // Left Rotation Speed

 int RightRotationSpeed = 250; // Right Rotation Speed


 void setup() {

   pinMode(in1,OUTPUT);

  pinMode(in2,OUTPUT);

  pinMode(in3,OUTPUT);

  pinMode(in4,OUTPUT);
```

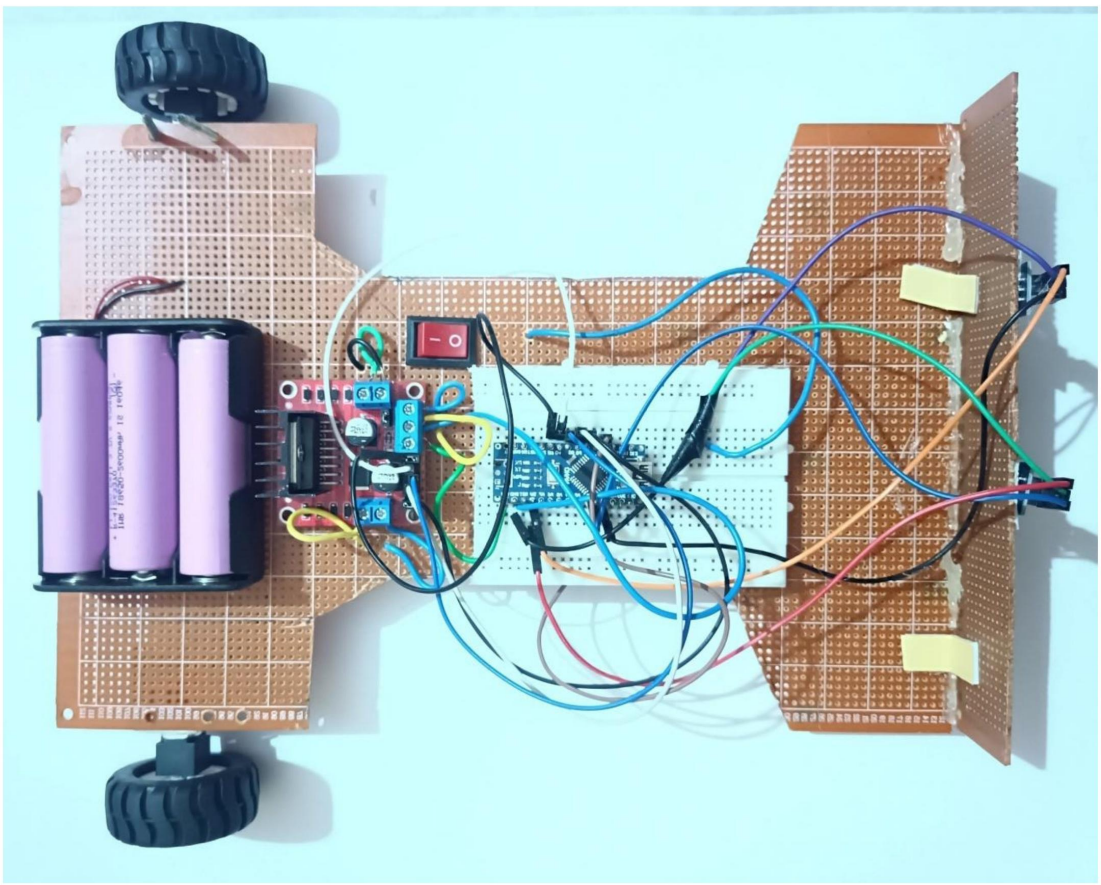
```
    pinMode(enA,OUTPUT);
    pinMode(enB,OUTPUT);

      pinMode(A1, INPUT); // initialize Left sensor as an input
      pinMode(A0, INPUT); // initialize Right sensor as an input

}

void loop() {

  int LEFT_SENSOR = digitalRead(A0);
  int RIGHT_SENSOR = digitalRead(A1);

if(RIGHT_SENSOR==0 && LEFT_SENSOR==0) {
    forward(); //FORWARD
}

  else if(RIGHT_SENSOR==0 && LEFT_SENSOR==1) {
    right(); //Move Right
  }
```

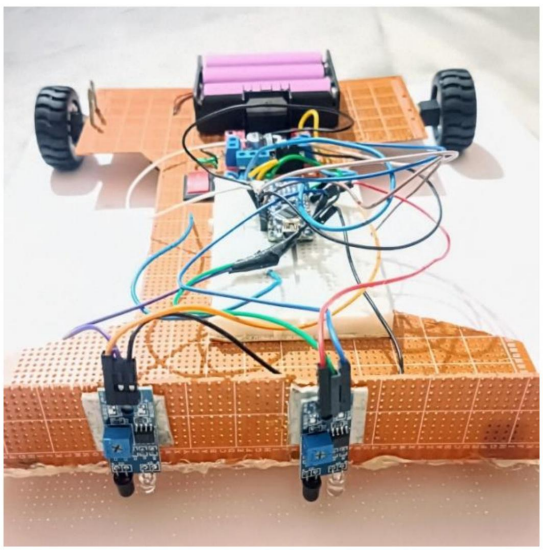
```
  else if(RIGHT_SENSOR==1 && LEFT_SENSOR==0) {
    left(); //Move Left
}


  else if(RIGHT_SENSOR==1 && LEFT_SENSOR==1) {
    Stop();  //STOP
 }
}
void forward()
{
          digitalWrite(in1, HIGH);
          digitalWrite(in2, LOW);
          digitalWrite(in3, HIGH);
          digitalWrite(in4, LOW);

             analogWrite(enA, M1_Speed);
             analogWrite(enB, M2_Speed);
}

void backward()
```

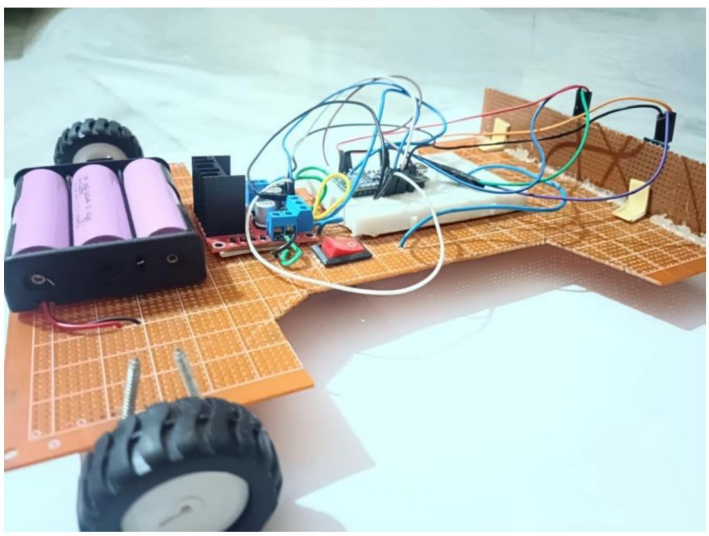
```
{
        digitalWrite(in1, LOW);
        digitalWrite(in2, HIGH);
        digitalWrite(in3, LOW);
        digitalWrite(in4, HIGH);

          analogWrite(enA, M1_Speed);
          analogWrite(enB, M2_Speed);
}

void right()
{
        digitalWrite(in1, LOW);
        digitalWrite(in2, HIGH);
        digitalWrite(in3, HIGH);
        digitalWrite(in4, LOW);

          analogWrite(enA, LeftRotationSpeed);
          analogWrite(enB, RightRotationSpeed);
}
```

```
void left()

{

        digitalWrite(in1, HIGH);

        digitalWrite(in2, LOW);

        digitalWrite(in3, LOW);

        digitalWrite(in4, HIGH);


           analogWrite(enA, LeftRotationSpeed);

           analogWrite(enB, RightRotationSpeed);

}


void Stop()

{

        digitalWrite(in1, LOW);

        digitalWrite(in2, LOW);

        digitalWrite(in3, LOW);

        digitalWrite(in4, LOW);

}
```

PROJECT PHOTOS







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NameofCourse from which Principles are applied in this project	**Related Course Outcome Number**	**Descriptionofthe application**	**Page Number**	**AttainedPO**

GuideSignature